

Opportunity to Grow, Incentive to Pursue: Future of U.S. Vocational-Technical Jobs in the Age of AI

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Executive Summary

In a 2026 Gallup survey, 79 percent of U.S. adults said they expect artificial intelligence to reduce the total number of American jobs over the next decade, up from 73 percent the year before. That fear echoes the anxiety that trailed every earlier wave of automation, from the Luddites who smashed power looms in the 1810s to the computerization that hollowed out middle-skill office and production jobs starting in the 1980s. For the vocational-technical trades, however, the best available evidence tells a more encouraging story. Occupations tied to Massachusetts' Chapter 74 vocational-technical programs, from electricians and HVAC technicians to medical assistants and restaurant cooks, make up just 12.3 percent of the nation's detailed occupations, yet they account for 16.0 percent of total U.S. employment, 15.2 percent of annual job openings, and 5.0 percent projected growth through 2034, nearly double the 2.7 percent projected for the rest of the labor market.

That advantage holds at every rung of the credential ladder and is widest at the bottom: VocTech occupations requiring no formal credential are projected to grow 7.3 percent, more than ten times the 0.7 percent rate for comparable traditional jobs, while also paying more on average. The pattern is not a Massachusetts artifact. Comparing the occupations that drive 80 percent of annual job openings in Texas, California, Florida, and Massachusetts shows the same trades leading growth nationally and within each state, even as the pace of growth varies by region.

The deeper question is whether artificial intelligence will do to these trades what computerization did to middle-skill work a generation ago: hollow out the middle while rewarding only the most credentialed and the least skilled. Using AI exposure scores from the leading large language model exposure framework, this report finds that many of the fastest-growing trades, cooks, HVAC mechanics, construction laborers, and electricians, sit well below the economy's average exposure to today's AI, while the most exposed occupations cluster in newer technical fields such as software development. Still, roughly ten of the twenty-five occupations examined sit above the average exposure line, enough to take seriously. The trades are not immune to AI, but neither are they its primary target.

To understand how career and technical education should respond, we surveyed 26 vocational-technical practitioners across Massachusetts, including superintendents, teachers, employers, advisory committee members, and students, and conducted in-depth interviews with four school leaders. Their perspectives, paired with national labor market data and the growing research literature on AI and CTE, ground five recommendations: embrace foundational skills more, not less; recognize that AI's impact differs sharply across and within trades; plan deliberately for the entry-level work AI may automate away; prepare students to work alongside AI where it augments rather than replaces judgment; and build both bigger and better career and technical education systems nationally. Vocational-technical education is not simply a hedge against an uncertain AI future. On the best available evidence, it may be one of the surer bets in the U.S. labor market.

1 Introduction

In 2017, eight researchers at Google published a paper with an unassuming title: “Attention Is All You Need” (Vaswani et al., 2017). It introduced the transformer, a new way for machines to process language. Most people never read it. Most people never heard of it. Five years later, everyone felt it.

On November 30, 2022, OpenAI released ChatGPT, a chatbot built on the transformer's architecture. It reached 100 million monthly users in about two months, faster than TikTok which took nine months or Instagram which took two and a half years to do the same (Hu, 2023). Google countered with Gemini. Anthropic answered with Claude. Meta released Llama. xAI introduced Grok. What began as one company's product launch became an industry-wide sprint.

In the last four years, artificial intelligence, especially large language models like ChatGPT and Claude, went from something few people had heard of to the thing you couldn't escape, on the news, in group chats, at the dinner table. The ground shifted under all of us, and it shifted fast.

That shift has not felt reassuring. In a 2026 Gallup survey, 79 percent of U.S. adults said they expect AI to reduce the total number of American jobs over the next decade, up from 73 percent the year before (Ray, 2026). The fear is real, and it is growing.

But why the fear? And is this fear tracking reality, or is it tracking headlines?

This fear around AI is historical. Every major innovation and consequent wave of automation has provoked the same anxiety, and more than once, that anxiety was justified. In the early 19th century during the Industrial revolution, English textile workers known as the Luddites smashed the power looms that were replacing their trade. Economic historian Joel Mokyr and his coauthors found that the hand-loom weavers and frame knitters of that era were, in fact, rapidly wiped out once factories took hold after 1815 (Mokyr et al., 2015, as cited in Autor, 2024). It took another five decades before working-class living standards recovered.

The next wave hit closer to our own era. Computers, unlike the machines of the Industrial Revolution, are good at one particular thing: executing routine tasks that follow explicit rules, like filing, calculating, and record-keeping (Autor et al., 2003). Starting in the 1980s, computerization automated away a broad stratum of middle-skill office and production jobs. It

complemented the work of highly educated professionals, whose judgment computers could not replace, and it left roughly 60 percent of American adults, those without a bachelor's degree, competing for a shrinking set of non-expert, low-paid service jobs (Autor, 2024). Economists call the result labor market polarization: growth at the top of the wage distribution, growth at the bottom, and a hollowed-out middle (Autor & Salomons, 2018; Dengler & Matthes, 2018; Katz et al., 2023; Kromann et al., 2020, as cited in Ho & Giwa, 2026). The fear now is that AI might repeat this story, only faster and wider, further exacerbating the inequality.

That fear is not unreasonable. But on the best available evidence, it does not track reality.

AI may not repeat this story, and the reason comes down to what computers were actually good at. Computers excelled at routine work: tasks with clear rules that could be programmed, such as filing records, running calculations, or searching databases. That made them powerful complements for professionals whose jobs still required human judgment. A physician could let software organize patient information, then spend more time on diagnosis, making their expertise more productive and better paid. The same technology, however, automated routine middle-skill jobs such as bookkeeping, payroll, and assembly-line work, eliminating many of them outright. Most displaced workers did not move into higher-paying professional occupations. Instead, they moved into service jobs like cooking, cleaning, and caregiving that required physical presence but little formal training, where the growing supply of labor kept wages low even as employment expanded. The result was labor market polarization: growth at the top of the wage distribution, growth at the bottom, and a hollowed-out middle (Autor et al., 2003; Autor & Dorn, 2013).

AI inverts this pattern. Computers were reliable at rules and useless at judgment. AI is closer to the opposite. It struggles with hard facts, but it is remarkably good at absorbing tacit knowledge. This ability allows it to take on one-off, high-stakes decisions, the kind of work that used to require an expert in the room (Autor, 2024). Autor's argument is that AI could let a much larger set of workers take on decision-heavy tasks that elite credentials used to gate: paralegals doing more of what lawyers do, technicians doing more of what engineers do, workers with a real foundation of training doing more of what used to require a much longer credential.

These changes raise fundamental questions about the future of work. Is the U.S. labor market signaling worker scarcity or surplus? Are vocational-technical occupations following the same trajectory as the rest of the economy, or are they different? Which trades are growing, which are shrinking, and how do these patterns vary across states? Finally, how might AI reshape (or not) the work performed in these occupations? Answering these questions provides a clearer picture of where vocational education is headed and what that means for policy, governance, and pedagogy.

We answer these questions by triangulating three sources of evidence. First, we analyze national data on labor market demand, occupational projections, and vocational-technical education enrollment. Second, we draw on the growing literature on AI, labor market change, and career and technical education. Third, we incorporate the perspectives of vocational-technical school leaders in Massachusetts, a state widely recognized for its regional vocational education model and strong evidence of positive student outcomes (Sinicola & Ferreira, 2022; Dougherty, 2018; Stevens et al., 2019).



Figure 1: Job Openings, Unemployment, and the Ratio Between Them

Note. Top panel shows job openings and the unemployment level in millions. Bottom panel shows the ratio of openings to unemployed people, with a dotted reference line at 1.0 (as many openings as unemployed people). Annual (December) snapshots through 2025, plus June 2026. Source: U.S. Bureau of Labor Statistics, Job Openings and Labor Turnover Survey and Current Population Survey, via FRED, Federal Reserve Bank of St. Louis.

2 Labor Market and the VocTech Advantage

The introduction closed with four questions about the future of work. This section takes up the first two: is the U.S. labor market signaling worker scarcity or worker surplus, and are vocational-technical occupations following the same trajectory as the rest of the economy, or moving on their own? Both questions matter beyond labor economics. As AI reshapes which skills the economy rewards, understanding where demand already exists, and who is positioned to meet it, will shape how the country harnesses AI to support productivity growth (Brynjolfsson et al., 2018; Filippucci et al., 2024; Ho & Giwa, 2026).

This section looks at two pieces of evidence to answer them. First, the balance between job openings and unemployed workers, a simple gauge of how much leverage jobseekers currently have. Second, how vocational-technical occupations compare to the rest of the labor market in growth, wages, and scale.

For most of the past twenty-five years, employers, not workers, held the advantage. From 2000 through 2017, the ratio of job openings to unemployed people stayed below 1.0, meaning

there were consistently more people looking for work than positions open. It fell as low as 0.17 in 2009 before briefly crossing 1.0 in 2018 and 2019, then surged to a record 1.89 in 2022 as employers scrambled to hire after the pandemic. Since then, it has cooled every year, reaching 0.87 in 2025 before ticking back to about 1.0 by June 2026.

That cooling is not necessarily bad news. The decline since 2022 reflects fewer job openings more than rising unemployment, suggesting employers are hiring more cautiously rather than shedding workers. The labor market today looks close to balanced, a better position for jobseekers than most of the 2000s and 2010s offered, even if it no longer favors workers as strongly as it did in 2021 and 2022.

That national picture, however, says nothing about where demand is concentrated. Answering that requires understanding vocational-technical education and the occupations it prepares students to enter.

Career and technical education, or CTE, is the modern term for what used to be called vocational education: structured training in fields like the skilled trades, healthcare support, and advanced manufacturing, delivered alongside or instead of a purely academic track. The field has changed considerably over the past two decades. Federal policy formally rebranded vocational education as “career and technical education” in 2006, explicitly to reduce stigma and to position it as a route to both college and career rather than a track for students who could not handle academic work. The most recent reauthorization of federal CTE law, the Perkins Act of 2018, pushed further, emphasizing high-quality work-based learning and aligning the U.S. system more closely with vocational models used internationally (Oskay & Plasman, 2026).

The evidence supports that shift. Students who complete CTE coursework in high school graduate at higher rates, attend more regularly, and are more likely to apply to and enroll in college, with effects often strongest for low-income students and students with disabilities (Bozick & Dalton, 2013; Dougherty, 2018; Plasman & Gottfried, 2018, 2021, 2023). The labor market evidence is equally encouraging. CTE participation has been linked to higher earnings, with positive effects documented in randomized evaluations, studies of California community colleges, and recent cost-benefit analyses of Massachusetts standalone CTE programs (Kemple & Willner, 2008; Plasman, 2019; Stevens et al., 2019; Dougherty & Smith, 2025). Increasing the amount of CTE coursework students take has also been associated with higher educational attainment and stronger labor market outcomes (Dougherty et al., 2019).

Autor (2024) argues that AI is more likely to expand the set of workers who can take on decision-heavy, judgment-based tasks than to replace them outright, extending exactly the kind of foundational, applied training CTE already provides. If that argument holds, the vocational-technical trades examined here are not waiting to be automated away. They may instead be one of the places AI complements rather than substitutes.

VocTech-destination occupations are a small slice of the labor market by headcount, just 12.3% of all detailed occupations (102 of 832), but they account for 16.0% of total U.S. employment and 15.2% of all annual job openings. They are also growing faster than everything else, 5.0% projected growth through 2034 compared with 2.7% for all other occupations combined.

That national picture is an optimistic one for vocational-technical trades, and it holds up once broken down by how much schooling a job typically requires. Table 2 splits the same

Table 1: Employment, Growth, and Openings, VocTech Destinations vs. Traditional Occupations

Track	Occupations (<i>n</i>)	Employment 2024	Employment 2034	Growth Rate	Annual Openings
VocTech destinations	102	27,263,400	28,614,900	+5.0%	2,857,600
Traditional / all other	730	142,693,000	146,555,000	+2.7%	16,004,000
All 832 occupations	832	169,956,400	175,169,900	+3.1%	18,861,600

Note. VocTech destinations = occupations mapped from Massachusetts Chapter 74 vocational-technical program list (102 of 832 detailed occupations). Traditional / all other = all remaining occupations (see Appendix A for details). Source: U.S. Bureau of Labor Statistics, Occupational Employment and Wage Statistics and Employment Projections, 2024–34.

Table 2: Employment, Growth, and Wages by Credential Tier, VocTech vs. Traditional Occupations

Credential Tier	Track	Occupations (<i>n</i>)	Employment 2024	Employment 2034	Growth Rate	Median Wage
No Formal Credential	VocTech	10	5,240,000	5,625,000	+7.3%	\$38,090
	Traditional	99	35,009,700	35,257,900	+0.7%	\$34,580
HS to Sub-Bachelor's	VocTech	72	16,065,000	16,635,000	+3.5%	\$48,620
	Traditional	334	53,414,400	54,290,200	+1.6%	\$46,040
Bachelor's or Higher	VocTech	20	5,958,400	6,354,900	+6.7%	\$80,690
	Traditional	209	32,140,500	33,881,300	+5.4%	\$81,680

Note. VocTech = occupations mapped from Massachusetts Chapter 74 vocational-technical program list (102 of 832 detailed occupations, see Appendix A for details). Traditional = all remaining occupations in that credential tier. Wages are median annual wage. Source: U.S. Bureau of Labor Statistics, Occupational Employment and Wage Statistics and Employment Projections, 2024–34.

comparison into three credential tiers: no formal credential, high school through sub-bachelor's, and bachelor's or higher.

VocTech occupations grow faster than their traditional counterparts at every credential level, and the signal is an optimistic one specifically for vocational-technical trades. The gap is largest at the bottom of the credential ladder: 7.3% projected growth for VocTech jobs requiring no formal credential, more than ten times the 0.7% rate for traditional jobs in that tier. It narrows further up the ladder, but never reverses: 3.5% versus 1.6% in the middle tier, and 6.7% versus 5.4% among jobs requiring a bachelor's degree.

Wages follow the same direction in the bottom two tiers. VocTech occupations pay more than their traditional counterparts among jobs with no formal credential (\$38,090 versus \$34,580) and jobs requiring a sub-bachelor's credential (\$48,620 versus \$46,040). Only at the bachelor's-or-higher tier does that edge disappear, where the VocTech median (\$80,690) runs almost identical to, and slightly below, the traditional-track median (\$81,680).

Together, these pieces of evidence answer indicate that vocational-technical trades are not simply following the rest of the economy's trajectory: they are a small share of all occupations that is growing faster, priced close to the rest of the labor market, and positioned to absorb a disproportionate share of the projected demand.

3 The VocTech Opportunity Landscape

The last section showed that vocational-technical occupations grow faster than the rest of the economy at every credential tier. This section asks two more specific questions: Which trades are growing fastest, and which are shrinking? And does that national picture hold across states with different economies and vocational education systems?

Growth alone does not tell the whole story. Figure 2 plots all 102 VocTech occupations by two measures: projected employment growth from 2024 to 2034 and annual job openings. Together, these capture both how fast an occupation is expected to grow and the scale of demand behind that growth. To highlight the strongest and weakest opportunities, the figure identifies the top 15 and bottom 15 occupations using an Opportunity Score. The score simply averages each occupation's percentile rank on growth and annual openings, giving equal weight to both measures. A detailed description of the methodology is provided in the codebook accompanying Appendix A.

Restaurant cooks top the ranking, combining 14.9% projected growth with roughly 250,700 annual openings, the highest in the dataset. Software developers post the fastest projected growth (15.8%) while still generating more than 115,000 openings each year. Medical assistants, electricians, and construction laborers round out the top five, each pairing strong growth with substantial labor demand. At the other end, occupations such as patternmakers, model makers, and prepress technicians are both shrinking and small to begin with, some generating only a few hundred openings nationwide each year. These are shrinking occupations in shrinking labor markets.

The strongest and weakest opportunities are distributed very differently across the credential ladder. The top 15 occupations are spread across all three education tiers, visible in Figure 2 as a mix of green, orange, and blue bubbles. Cooks and construction laborers require no formal credential, medical assistants and electricians sit in the middle tier, and software developers require a bachelor's degree. The bottom 15 tell a different story. They are almost entirely orange, clustering in the high school to sub-bachelor's tier. Most also share a common type of work: routine production and manufacturing tasks such as patternmaking, model making, and metal and plastic tooling. As discussed in the Introduction, Autor (2024) argues that AI is most likely to complement work built around tacit judgment while substituting for tasks that are routine and standardized. The pattern in Figure 2 is consistent with that argument. The middle credential tier is the most divided. It contains both some of the strongest opportunities and nearly all of the occupations in decline. What might separate them is not education, but the nature of the work itself.

National averages can hide important regional differences. To see how much, we first identified the occupations that together accounted for 80% of annual job openings in the United States and in each of our four case-study states: Texas, California, Florida, and Massachusetts. We then pooled those five lists and removed duplicates, producing a final set of 25 occupations that captures the vast majority of labor demand across these geographies. The four states were chosen because they represent different regions, economies, and approaches to career and technical education. Figure 3 compares projected employment growth across those occupations.

The results are encouraging. Roughly four out of five occupations grow faster than the U.S. economy-wide average of 3.1%, most by a wide margin. Some patterns are remarkably



Figure 2: The 102 VocTech Destination Occupations: Growth vs. Openings, Ranked by Opportunity Score

Note. Bubble size represents total employment; color represents an occupation's typical entry-level education tier (no formal credential, high school through sub-bachelor's, or bachelor's or higher). Labeled occupations are the top 15 and bottom 15 of the 102 VocTech occupations by Opportunity Score, an average of each occupation's percentile rank on projected employment growth (2024-34) and projected annual openings (2024-34). Source: U.S. Bureau of Labor Statistics, Employment Projections, 2024-34.

consistent. Restaurant cooks post double-digit growth in every state, ranging from 14.3% in Massachusetts to 26.3% in Texas. Others vary much more. Software developers grow 15.8% nationally, but projections range from about 8% in Massachusetts to nearly 29% in Texas. Farmworkers, teaching assistants, and elementary teachers are the only occupations projected to decline nationally, although even farmworkers grow modestly in Texas and California. National projections are a useful starting point. Local projections are often the better planning tool.

The purpose of Figure 3 is not to compare states against one another. It is to test whether the national story holds. It does. The occupations leading nationally, software developers, restaurant cooks, medical assistants, electricians, HVAC mechanics, and construction laborers, also rank among the strongest opportunities across Texas, California, Florida, and Massachusetts. States differ mainly in magnitude. Texas and Florida generally project faster growth, while Massachusetts tends to be more conservative. But the overall ranking changes very little. That consistency matters. It suggests the opportunities identified in this report are not unique to one state, but reflect a broader national pattern.

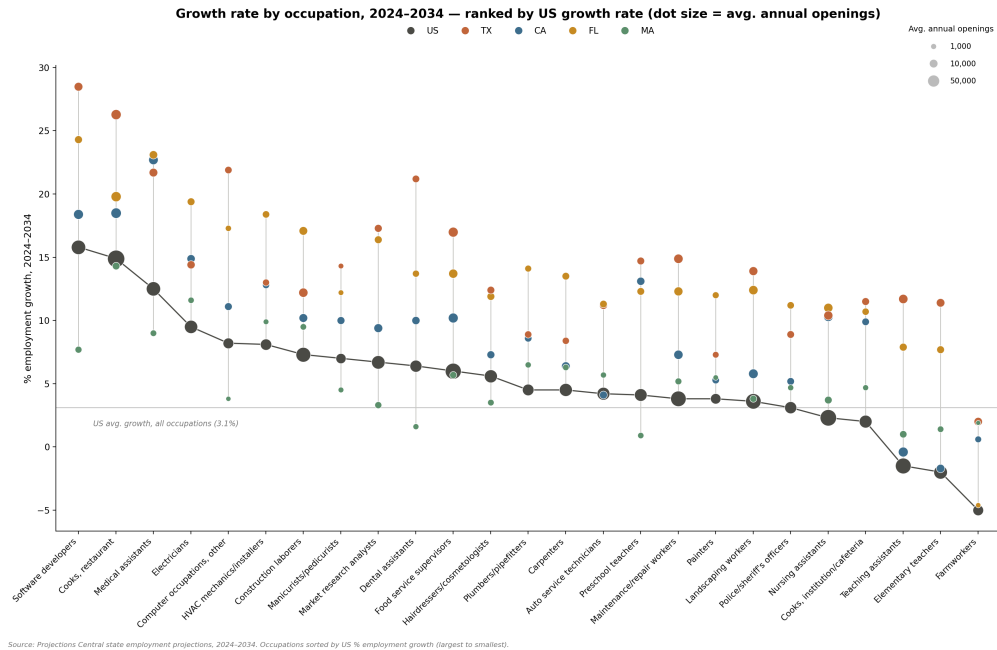


Figure 3: Growth Rate by Occupation, 2024-2034, Ranked by U.S. Growth Rate (Dot Size = Average Annual Openings)

Note. Dot size represents average annual openings. Occupations are sorted by U.S. percent employment growth, largest to smallest; the dashed line marks the economy-wide U.S. average growth rate across all occupations (3.1%). The 25 occupations shown are those that account for at least 80% of annual openings in the U.S. as a whole, or in at least one of the four states shown, pooled together (see text). Source: Projections Central, state occupational employment projections, 2024-2034.

4 AI Exposure on VocTech Occupations

The Introduction closed with a simple question: are trade jobs really immune to AI? This section examines what the evidence actually shows.

On one side, Elon Musk has called AI “the most disruptive force in history” and predicted a point where “no job is needed” (Henshall, 2023). Geoffrey Hinton, often described as the Godfather of AI, has suggested that plumbing may be one of the safest careers left (Boran, 2023). On the other side, OpenAI CEO Sam Altman has argued that investors are “overexcited about a kernel of truth” (Nolan, 2025). Veteran investor Charlie Munger has been equally skeptical, calling the current AI moment “more hype than it deserves” (Mancini, 2023). Two popular camps. Both confident.

What does the evidence suggest?

Nearly three years after ChatGPT’s release, a recent Yale Budget Lab analysis found no discernible disruption yet to the broader U.S. labor market, undercutting the loudest fears that AI is already eroding demand for cognitive labor at scale (Gimbel et al., 2025). But not every piece of evidence points in the same direction. McKinsey projected AI could add \$13 trillion to global economic output by 2030 (McKinsey Global Institute, 2018), and Harvard economist

Jason Furman found that AI-related investment, just 4 percent of U.S. GDP, accounted for 92 percent of all U.S. GDP growth in the first half of 2025 (Lichtenberg, 2025). The picture is not settled. It is mixed.

Answering that question begins with one distinction economists return to repeatedly: automation versus augmentation. Automation replaces a task by doing the work a person used to do. Augmentation supports a task, making the person faster, better, or more capable. AI can do either, depending on the job and how it is used (Autor, 2022; Zarifhonarvar, 2024). As Autor (2024) puts it, a nail gun does nothing for someone who has never framed a wall, but it allows a trained roofer to work much faster.

We next asked how the current wave of AI, the large language models behind ChatGPT and similar systems, actually relates to the trades. The best available evidence comes from Eloundou and colleagues (2024), who developed the leading exposure framework for large language models. Using federal O*NET task data rated by both human annotators and GPT-4, they estimated how exposed different occupations are to this technology. When our team reached out, they generously shared the underlying data, allowing us to calculate AI exposure scores for the 25 key VocTech occupations examined in this report.

O*NET classifies occupations at a finer level than the BLS system used elsewhere in this report, so several O*NET occupations often mapped to a single occupation in our dataset. In those cases, we averaged the AI exposure scores to produce one score for each of the 25 occupations. Scores ranged from 0.02 for landscaping workers to 0.87 for software developers. For ease of interpretation, we grouped them into three exposure tiers: low, medium, and high.

Low AI exposure does not mean AI has nothing to offer these occupations. It suggests that the core tasks in these jobs are currently less exposed to substitution by AI. That brings us back to the distinction between automation and augmentation introduced earlier. For many trade occupations, AI may become a tool that helps workers perform their jobs better rather than replacing them, consistent with the pattern Autor (2024) argues we should expect.

Three recent studies point the same direction. Noy and Zhang (2023) gave professional writers access to ChatGPT and found time on task fell 40 percent, with the largest quality gains concentrated among the weakest writers. Peng et al. (2023) found programmers using GitHub Copilot completed a coding task 56 percent faster than those without it. Brynjolfsson, Li, and Raymond (2023) found customer service agents using an AI assistant resolved 15 percent more issues per hour on average, with the largest gains among newer, less experienced agents and small quality declines among the most experienced ones. In each case, AI did not replace the expert at the top. It closed the gap between the experts and non-experts.

This comes with a caveat worth taking seriously. Autor and Thompson (2025), in a more recent and more technical paper than the one anchoring this report's introduction, argue that an occupation's fate depends on which tasks automation removes, not just how exposed the occupation looks overall. If AI strips out a job's easiest tasks, what remains demands more expertise: wages rise, but fewer workers qualify, so employment falls. If it strips out the hardest tasks, the opposite happens: wages fall, but employment rises because more people can now do the job. A trade sitting in the low-exposure band today is not guaranteed to stay there, and low exposure alone does not say which of these two paths a given occupation is on.

Put together, trades are not immune to AI. Roughly ten of the twenty-five occupations examined

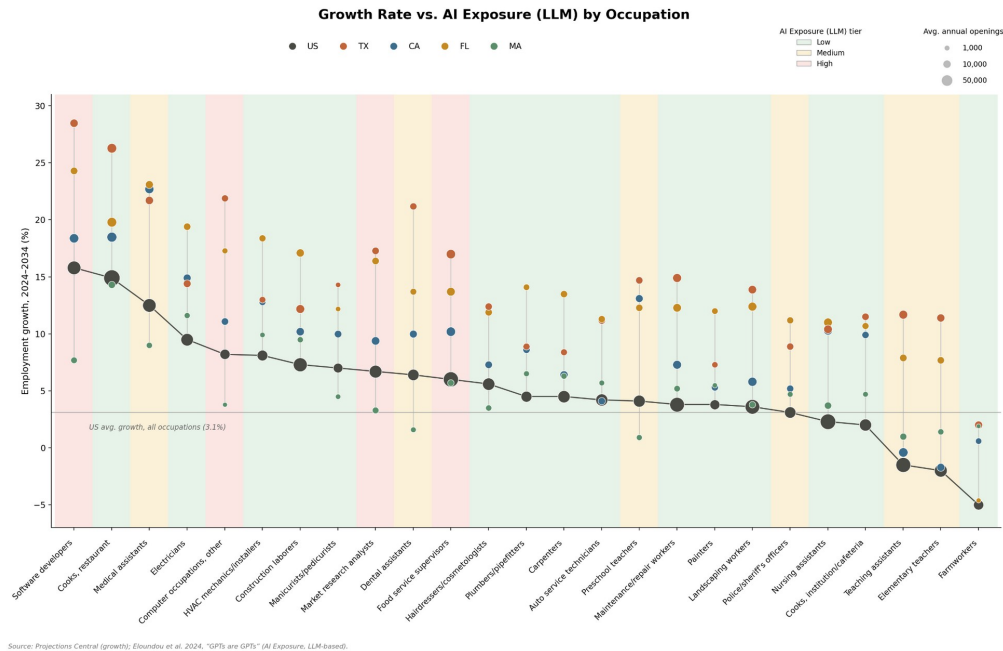


Figure 4: Figure 4. Growth Rate vs. AI Exposure (LLM) by Occupation

Note. AI exposure scores are adapted from Eloundou et al. (2024) and aggregated to the 25 VocTech occupations using the O*NET to SOC crosswalk. Occupations scoring at or below the economy-wide average exposure (0.15) are classified as low, those between 0.2 and 0.4 as medium, and those at 0.5 or above as high. Dot size represents average annual openings; dot color represents state. Sources: Projections Central, state occupational employment projections, 2024 to 2034; Eloundou et al. (2024).

here sit above the average AI exposure level, enough to take seriously. But neither are the trades the primary targets of the current wave of large language models. The occupations with the highest exposure are concentrated in newer technical fields such as software development, computer occupations, engineering, robotics, and digital design, while many of the fastest-growing hands-on trades remain below the average exposure line. The evidence therefore aligns closely with the framework developed in the Introduction. AI appears more likely to complement work built around physical skill and tacit judgment than to replace it.

The evidence so far points in a broadly encouraging direction. VocTech occupations account for a disproportionately large share of labor demand, continue to grow across states, and many remain relatively less exposed to the current generation of AI. But these are still structural patterns. They cannot tell us what is happening inside classrooms, workshops, or clinics, or how educators and employers are actually responding to AI. For that, we turn to the people leading vocational education.

5 Implications and Recommendations

Massachusetts is a good place to ask what these findings mean in practice. Its regional vocational-technical schools are among the most established and widely studied in the country, with strong evidence of positive effects on graduation, earnings, and long-term employment (Sinicola & Ferreira, 2022; Dougherty, 2018; Stevens et al., 2019; Feinberg et al., 2025). Our

own earlier work reached a similar conclusion from the perspective of practitioners (Raj et al., 2026). To understand how career and technical education should respond to AI and adapt to a changing labor market, we turned to the people who run these schools, teach in them, and hire their graduates.

We surveyed 26 practitioners across Massachusetts, including superintendents, teachers, employers, advisory committee members, and student representatives. Four also participated in in-depth interviews: Karen Maguire, superintendent of Tri-County Regional Vocational Technical High School; Heidi Driscoll, superintendent of Minuteman Regional Vocational Technical High School; John Genereux, vice president of Hamilton Storage, industry advisor, and school committee member at Tri-County; and Robert Sanborn, superintendent of Cape Cod Regional Technical High School. Their perspectives, together with the evidence presented throughout this report, form the basis for the recommendations that follow. We also thank Heidi Driscoll and Alice DeLuca for welcoming us to Minuteman, where seeing the shops firsthand grounded many of the ideas developed here.

Throughout the responses, one message came through before anything else: AI is moving faster than the systems built to respond to it. As Robert Sanborn put it, the technology is advancing “much faster than they can write the frameworks.” The recommendations that follow are an attempt to close that gap.

5.1 Embrace Foundational Skills More, Not Less

Autor’s central argument is that AI changes access to expertise and high-stakes decision making, but it does not eliminate the need for foundational knowledge and training. AI can help workers exercise judgment that once required years of experience, but only when they already have a strong foundation to build on (Autor, 2024). The same principle applies to career and technical education. As Table 2 showed, many VocTech occupations that require little or no formal credential are projected to grow rapidly while offering strong wages. Succeeding in those occupations, however, still depends on building real, hands-on skills. AI can extend what that training can achieve. It cannot replace it.

The practitioners we spoke with drew the same distinction between exposure and mastery. Heidi Driscoll cautioned against “vocational light” programs that give students a taste of a trade without truly preparing them for it. “Exposure is very different from training,” she told us. Karen Maguire made a similar point about AI. Even if AI performs a task such as diagnostics, students still need to understand the fundamentals so they can judge whether the AI is right or wrong. John Genereux, who hires technicians directly, agreed that the risk lies less in the tool than in the person using it. “If you don’t know what to ask it, it will give you the wrong answer,” he told us. “If you feed it garbage, you’re going to get garbage.” Dan Charbonneau, a baking instructor at Minuteman, extended that idea to his own trade, warning that AI “diminishes creativity in culinary arts” when it substitutes for, rather than supports, students’ own thinking. Erik Synnestvedt, who teaches computer science and cybersecurity at Minuteman, described the practical response: a senior-year module that teaches students “how to use AI responsibly to boost their career prospects and performance, not to replace themselves.” Different practitioners. The same message. Build the foundation first. Then teach students how to use AI on top of it. Responsibly.

5.2 Understand AI's Differential Impact

AI does not affect every trade equally. Figure 4 showed that many traditional hands-on trades, such as construction, electrical, and HVAC, remain relatively less exposed to AI than newer technical fields such as computer science, engineering, and design. Similar patterns have been reported in previous research (Webb, 2020; Zarifhonarvar, 2024). Career and technical education should respond accordingly. A one-size-fits-all AI strategy will not work.

Practitioners are already seeing these differences. Several respondents expected computer science and design and visual communications to change substantially as AI advances. Robert Sanborn, for example, expects design and visual communications to evolve drastically over time. Heidi Driscoll similarly identified computer science, engineering, art, and design as the trades most likely to feel AI's effects first, while expecting hands-on trades to be less affected. Karen Maguire cautioned against thinking the building trades are immune, arguing instead that AI will reshape diagnostics, predictive maintenance, and building automation while leaving hands-on work central.

The differences extend beyond individual trades to the work students do within them. John Genereux, who also serves as an industry advisor for the Computer Information Systems program, noted that students in the same program often prepare for "two different futures": networking and cybersecurity or software development. Figure 2 hints at the same pattern. Computer occupations appear several times among the highest opportunity scores, reflecting the different career pathways that exist within a single vocational program. AI exposure follows a similar pattern. As discussed earlier, the *GPTs are GPTs* paper (Eloundou et al., 2023) showed that software development (about 0.8) is among the most exposed computing occupations, while networking and cybersecurity (about 0.6 each) remain comparatively less exposed, even though all fall within the broader computer science field. Career and technical education should reflect that level of nuance. School leaders, curriculum developers, program advisors, instructors, and AI experts will need to work together to understand how AI is reshaping individual trades and prepare students accordingly.

5.3 Plan for AI-Driven Automation

History shows these transitions require more than changes in the classroom. As Autor (2024) argues, expanding the role of nurse practitioners required new educational pathways, certification systems, and changes to professional regulations (Asubonteng et al., 1995). It also required health information technology robust enough to let nurse practitioners make and communicate high-stakes decisions on their own (Li et al., 2012). Only then could they take on responsibilities traditionally reserved for physicians. AI may require a similar institutional response. States should work with employers, industry partners, and licensing bodies to identify where AI is beginning to automate entry-level work and develop new pathways before those opportunities disappear. Co-ops, apprenticeships, internships, and structured on-the-job training may become even more important as traditional first jobs evolve. The goal is not to resist automation. It is to build the safeguards and institutions needed to help students adapt to it.

5.4 Prepare for AI-Driven Augmentation

AI will automate some tasks for some occupations, but it will augment many tasks across many occupations. As Autor and Thompson (2025) argue, the impact of automation depends on which tasks disappear and which remain. As routine and procedural work becomes automated, the remaining work often demands more judgment, problem solving, and expertise.

Practitioners described exactly this shift. John Genereux, who works in automation, drew a sharp distinction between repetitive work and what he called “actual thought and infinite mobility.” Robots can already handle much of the former, he argued, while electricians, HVAC technicians, and carpenters still work in changing physical environments where AI is more likely to become a tool than a substitute. Many practitioners cautioned against assuming that this means the trades are immune. Karen Maguire put it simply: “We keep hearing things like AI can’t touch the building trades.” That assumption, she argued, misses how AI is already entering those occupations through diagnostics, predictive maintenance, smart buildings, and digital planning. Robert Sanborn hopes AI becomes “a job changer” rather than “a job replacer,” freeing workers to spend more time on the “creative and higher value aspects” of their jobs instead of routine tasks.

Augmentation also creates new work. Autor (2024) points to occupations such as wind turbine technicians that emerged as new technologies created new industries. AI is beginning to do the same. One manufacturing employer in our survey reported that “electronics is booming due to AI demand” and that finding enough qualified workers has become the real challenge. Career and technical education should therefore prepare students not only for today’s occupations, but also for how those occupations will evolve and the new ones that will emerge. A growing body of research suggests that many of the fastest growing occupations will not involve humans or AI working alone, but humans working effectively with AI (Autor & Thompson, 2025; Ho & Giwa, 2026; Filippucci et al., 2024). The goal is to prepare students for that future.

5.5 Build Bigger and Better

Meeting the labor demand documented in this report will require both more career and technical education and better career and technical education. States already offer different models for doing this. Texas spreads CTE broadly through comprehensive high schools, California embeds career pathways through Linked Learning, and Florida uses career academies tied to industry certification. Massachusetts goes deeper through regional Chapter 74 schools, where students spend four years moving between academics and intensive shop training. Each model solves a different part of the problem. Breadth brings more students in. Depth prepares them further. The national challenge is to find ways to do both.

Massachusetts shows what that depth requires, and where it can break down. Robert Sanborn described employer advisory committees attached to every program, helping schools decide what to teach and what equipment students need. Karen Maguire reported that at least half of Tri County’s grade 12 students now participate in co-op placements with industry partners. John Genereux described the result more broadly: vocational education teaches students “how to basically be a young adult,” not just how to perform a trade. But this ecosystem is expensive and difficult to scale. Sanborn’s school received 440 applications for only 190 seats, and many

of the other schools we spoke with described similar waitlists and capacity constraints. Building bigger therefore means more schools, programs, seats, instructors, and employer partnerships. Building better means protecting the relationships and training quality that make those seats valuable in the first place.

The deeper challenge is cultural. For decades, American education has treated the four year degree as the clearest signal of ability and vocational education as something less. That perception has begun to change. Nationally, successive Perkins reauthorizations have deliberately repositioned vocational education as career and technical education, emphasizing college readiness, technical skills, and work based learning rather than remedial tracking (Gordon & Schultz, 2020; Oskay & Plasman, 2026).

The labor market is also increasingly reinforcing that shift. Employers are placing greater value on demonstrated skills, particularly in fast changing fields where formal credentials cannot keep pace. Bone et al. (2025), analyzing roughly eleven million job postings, found growing skill based hiring in AI roles alongside declining demand for university credentials and substantial wage premiums for AI skills. The implication reaches well beyond AI. Apprenticeships, vocational schools, community colleges, on the job training, and other skill building pathways deserve the same seriousness we have traditionally reserved for degrees. Career and technical education should keep raising its standards, expanding its reach, and rebuilding its image around what ultimately matters: what people can do. A country facing shortages of electricians, technicians, healthcare workers, builders, and other skilled professionals cannot afford to treat the people doing that work any less. They are part of the skilled workforce on which everyone else depends.

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Appendix A: Data and Matching Methodology

The analyses in this report begin with the 832 detailed (“line-item”) occupations reported in the U.S. Bureau of Labor Statistics (BLS) Employment Projections 2024–2034 dataset. Summary occupations were excluded because they do not report employment, wage, growth, or job opening data.

The 46 Massachusetts Chapter 74 vocational-technical programs were then matched to occupations using the official NCES/BLS CIP-to-SOC Crosswalk. Occupations that were clearly not realistic entry-level destinations, such as postsecondary teaching positions and managerial or first-line supervisory roles, were removed.

Most programs (40 of 46) were identified directly through the crosswalk. Six programs required manual review because they had no suitable crosswalk match or the listed occupation was not an appropriate destination. In those cases, occupations were assigned based on the Chapter 74 program curriculum. A small number of outdated SOC codes were also updated to match the 2024–2034 BLS classification.

Some occupations that typically require a bachelor’s degree or higher were retained because they represent recognized employment pathways for graduates of certain Chapter 74 programs in Massachusetts.

After removing duplicate matches across programs, the final list contained 102 unique occupations, referred to throughout this report as the VocTech occupations. Employment, projected growth, annual openings, wages, and educational requirements for these occupations were obtained from the BLS Employment Projections 2024–2034 dataset.